RAJAGANAPATHY M

AI/ML Engineer | LLM Engineer | Agentic AI | RAG | Computer Vision
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2+ Years Independent AI/ML Engineer (2023 – Present) | **M.Tech AI (CGPA 9.6/10, SRM Institute, May 2026)** | **Indian Patent Filed** (App No. 202641043947) | **IEEE Paper** Published | **1 Live Production Product Shipped to a Paying Client** (Anbu Health AI) | 3 Live HuggingFace Models | AWS + Azure + GCP | Prior 9+ years Electrical Engineering (2014–2023) | **Immediate Joiner**

Technical Skills

GenAI / LLMs	LLM Orchestration, RAG, FAISS, Qdrant, LangChain, HuggingFace, QLoRA, PEFT, LoRA, Agentic Workflows
Deep Learning	PyTorch, TensorFlow, Transformers, CNN, LSTM, BLIP-3, CLIP, ViT, Qwen, CodeLlama, TinyLLaMA
ML / NLP	scikit-learn, XGBoost, Gradient Boosting, SMOTE, TF-IDF, K-Means, NLTK, TextBlob, spaCy
Cloud / MLOps	AWS (SageMaker, S3, EC2), Azure (Container Apps, OpenAI), GCP (Vertex AI), Docker, GitHub Actions, FastAPI
Data Engineering	pandas, NumPy, MySQL, MongoDB, PostgreSQL, Supabase, ETL Pipelines, SQLAlchemy
Languages	Python, SQL, R, HTML, Bash
Security Compliance	DPDP Act, PHI redaction, consent audit logging, rate limiting, Cyber Security specialization (4 courses, 54.5 hrs)

Key Projects

P.1 — Anbu Health AI (Live Production Product, Paid Client) *React + FastAPI | Azure Container Apps | Groq + GPT-4o Vision | Qdrant RAG*

- Bilingual (Tamil/English) medical AI assistant analyzing lab reports, scans & medicine photos — live at anbuclinic.me, shipped to a real client under AI Vision
- 5-stage reasoning pipeline (routing → RAG → LLM → confidence scoring → hallucination check); Firebase OTP auth, DPDP Act 2023-aligned data handling

P.2 — Antahkarana Reasoning Framework (Patent filed + IEEE Paper) *Patent Filed App No. 202641043947 | IEEE Paper*

- Cognitively-inspired 5-stage reasoning architecture for LLMs & VLMs (manas → chitta → buddhi → ahamkara → sakshi); validated on 2,500+ multimodal samples
- Powers Anbu Health AI in production | Built with Transformers, BLIP-3, Qwen, LangChain, FAISS/Qdrant

P.3 — CodeLlama Coding Assistant (Applied Fine-Tuning) *AWS SageMaker ml.g4dn.xlarge | QLoRA | PEFT | HuggingFace Spaces*

- Fine-tuned CodeLlama-7B on CodeAlpaca-20K with **60% GPU memory reduction** via 4-bit quantization
- Deployed live demo on HuggingFace Spaces with Groq API inference

Professional Experience

Founder & Independent AI/ML Engineer — Jul 2023 – Present (2+ Years)

AI Vision (Self-Employed, MSME Registered — UDYAM-TN-02-0483528)

- Designed and shipped **Anbu Health AI** end-to-end (frontend, backend, AI pipeline, cloud infra) to a paying client — live in production on Azure
- Filed 1 Indian patent, IEEE paper, and published 3 fine-tuned models on HuggingFace — independent, project-based work, not a salaried role

Electrical Construction Site Engineer — Jan 2017 – Jun 2023 (6.5 Years)

SR Electrical Works, Chennai

- Sole engineer across 6 landmark projects (2 hospitals, 2 hotels, 1 medical college, 1 engineering college), Rs. 5–6 Cr portfolio, 20 labourers; zero site rejections under IEC/NBC codes.
- Prepared electrical shop drawings, quantity calculations and BOQs, and supported site execution for building and multi-specialty hospital projects.
- Explored AI/ML, computer vision, OCR and CAD automation to detect electrical symbols, automate quantity take-offs and extract cable measurements from drawings, the groundwork for my move into AI.
- Drove **10× profit growth** YoY by restructuring procurement & billing during the 2017 GST transition.

Electrical Maintenance Engineer — Dec 2014 – Dec 2016 (2 Years)

Mod Forge Pvt Ltd, Chennai (ISO/TS 16949)

- Maintained 500 kVA Site, 250 kVA DG and motors up to 400 kW on CNC/VMC lines with **zero unplanned downtime** over 2 years.

Education

M.Tech — Artificial Intelligence — SRM Institute of Science & Technology, Kattankulathur — Jun 2024 – June 2026

IITM Pravartak Certified AI & ML Program — GUVI / IIT Madras + NASSCOM — Nov 2023 – Nov 2024

B.E — Electrical & Electronics Engineering — Thangavelu Engineering College, Chennai — 2013

Achievements & Certifications

- **Indian Patent Filed** — App No. 202641043947 — Apr 2026
- **IEEE Paper Published** — Google Scholar Indexed
- **1 Product Live in Production** for a paying client (Anbu Health AI)
- **NPTEL TOP 5%** — IoT — 86% / 50,282 candidates (Oct 2025)
- **3 Live HuggingFace Models** — TinyLLaMA + Mini-VLM + CodeLlama-7B QLoRA
- **AWS Solutions Architect Associate** (Oct 2025)
- **Kaggle AI Agents Intensive** with Google (Dec 2025)
- **Hackathon 6th Place** / 195 — SRM (Dec 2024)